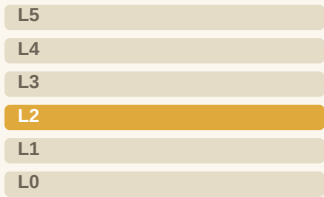


OPEN-SOURCE MODEL ASSESSMENT

DeepSeek V3



L2 · Open weights under a permissive license

DeepSeek AI · Released Not available · Prepared for Non-profit / public sector · Report date 2026-07-16

EXECUTIVE SUMMARY

The MIT license permits commercial use, modification, and redistribution, requiring only retention of the copyright and license notice; this satisfies the criteria for L2 (open weights under a permissive license). For a non-profit / public sector reader, DIT recommends: adopt with caution.

OUR RECOMMENDATION

ADOPT WITH CAUTION

OPENNESS GRADE

L2 · Open weights under a permissive license

LICENSE POSTURE

Commercial use permitted

DECISION AT A GLANCE

PRIVATE

ADOPT WITH CAUTION

ENTERPRISE

ADOPT

NON-PROFIT

ADOPT WITH CAUTION

01 Openness Ladder verdict

L2 · Open weights under a permissive license

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PRIMARY SOURCE

[P1] MIT (P1) <https://github.com/deepseek-ai/DeepSeek-V3/blob/main/LICENSE>

02 License summary

YOU ARE ALLOWED TO

- Commercial use
- Modification
- Redistribution

YOU ARE NOT ALLOWED TO

- Remove the MIT license or copyright notice

COMMERCIAL USE

MIT permits commercial use, modification and redistribution provided the copyright and license notice are retained.

03 Deployment and technical profile

MINIMUM HARDWARE

~400 GB VRAM (Q4_K_M) requiring multi GPU/distributed setup

RUNS FULLY OFFLINE

Yes

DEPLOYMENT VERDICT

needs a small server

TECHNICAL DETAIL

CONTEXT LENGTH

128K tokens

PARAMETERS

671B total / 37B active (MoE, 256 experts)

DISTRIBUTION FORMATS

Safetensors, GGUF

PRECISION

BF16

QUANTIZATION IMPACT ON VRAM

Moving from BF16 ("H1.34 TB VRAM for the full 671B parameter dense equivalent) to GGUF Q4_K_M reduces VRAM to ~400 GB, a ~70% reduction that enables multi GPU deployment.

MINIMUM VRAM

~400 GB (Q4_K_M)

VRAM BY VARIANT (MIN, QUANTIZED)

DeepSeek-V3 ~400 GB (Q4_K_M)

MODEL SERIES

Variant	Parameters	Min VRAM	Quant options	Notes
DeepSeek-V3	671B total / 37B active (MoE)	~400 GB (Q4_K_M)	Q4_K_M, Q8_0	Multi-GPU / distributed
DeepSeek-V3 (distilled)	varies by distillation	Not available	GGUF	Community distills

04 Decision frames

PRIVATE / INDIVIDUAL

The full DeepSeek-V3 model requires ~400 GB VRAM in Q4_K_M quantization, which exceeds the capacity of a single consumer GPU, making local deployment impractical for most individuals without access to multi GPU servers or cloud instances.

ADOPT WITH CAUTION

ENTERPRISE

Enterprises can meet the hardware requirements with multi GPU servers or cloud instances; the MIT license permits commercial use, modification, and redistribution with only attribution required.

ADOPT

NON-PROFIT / PUBLIC

Nonprofits with access to sufficient GPU resources (e.g., via research grants or cloud credits) can deploy the model under the permissive MIT license; otherwise, smaller community distilled versions may be more feasible.

ADOPT WITH CAUTION

05 Where sources disagree

Some sources (e.g., source 4, 7, 9) discuss DeepSeek V3.2 or V3.5 specifications (VRAM ~1.4 TB) which pertain to a different version and should not be conflated with DeepSeek V3.

06 Sources and attribution

This report was synthesized with the DIT OpenModel Synthesis Framework (OMSF), a practical, buyer-side adaptation of the Model Openness Framework (LF AI and Data Foundation, 2024; arXiv:2403.13784) and the OSI Open Source AI Definition. Figures are sourced from the web at report date and reflect publicly available information only.

- [P1] 1. DeepSeek-V3 model card (Hugging Face) <https://huggingface.co/deepseek-ai/DeepSeek-V3> · Not available
- [P1] 2. DeepSeek-V3 GitHub repository <https://github.com/deepseek-ai/DeepSeek-V3> · Not available
- [P1] 3. Unsloth DeepSeek-V3 GGUF <https://huggingface.co/unsloth/DeepSeek-V3-GGUF> · Not available
- [P1] 4. DeepSeek-V3 Technical Report <https://arxiv.org/html/2412.19437v1> · Not available